

Neel Vittal Bharath Vadlamudi

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Education

University of Maryland, Baltimore County

M.P.S. in Data Science, GPA: 3.67

Baltimore, MD

May 2026

National Institute of Technology Delhi

B.Tech in Computer Science and Engineering

Delhi, India

May 2023

Skills

Analytics & Data: Python, SQL, R, SAS, Stata, PySpark, Apache Spark, Azure Databricks, Pandas, NumPy, GeoPandas

Machine Learning & AI: PyTorch, HuggingFace, BERT, GPT, T5, LLM evaluation, RAG, ensemble ML, A/B testing, causal inference

Business Intelligence & Engineering: Power BI, Tableau, Flask, REST APIs, Streamlit, Git, CI/CD, AWS S3/Lambda/SQS

Risk & Quant Methods: Monte Carlo simulation, VaR/CVaR, reconciliation analytics, regulatory reporting, data quality controls

Experience

AI Institute of South Carolina

Data Infrastructure Research Assistant

May 2025 - Aug 2025

Columbia, SC

- Built Azure Databricks pipelines for 2.3M+ records across 5 state agencies, supporting government-partnered research with 99.7% data availability.
- Reduced data preparation time by 70% by standardizing ingestion, validation, and transformation for multi-agency datasets.
- Partnered with technical and nontechnical stakeholders to improve data quality checks, document lineage, and research usability.

UMBC + Meta AI Research

Machine Learning Research Assistant

Dec 2024 - Feb 2025

Baltimore, MD

- Improved model accuracy by 15% across 12+ benchmark datasets through transformer fine-tuning with PyTorch, HuggingFace, BERT, GPT, and T5.
- Designed a formal LLM evaluation methodology that surfaced 3 previously undetected bias issues in benchmark model behavior.
- Ran controlled benchmark experiments and translated findings into concise research summaries for faculty and technical collaborators.

HDFC Bank

Business Analyst / Project Lead

Jun 2023 - Jul 2024

Mumbai, India

- Prevented \$2.1M in operational risk exposure by modeling 1M+ Monte Carlo scenarios and improving banking controls.
- Reduced reconciliation effort by 60% across 500K+ daily records from 8 source systems, eliminating 12 discrepancy tickets per quarter.
- Automated recurring reporting and exception workflows, saving 15+ analyst-hours per week across 4 business lines.
- Promoted mid-engagement to Project Lead for CLMS, coordinating 6+ legal, compliance, vendor, QA, operations, and IT teams.

Projects

CDC Epidemiological Intelligence Platform

RAG, LLMs, LSTM, Spark

- Built a public-health signal workflow combining retrieval-augmented generation, sequence modeling, and Spark processing.
- Structured epidemiological outputs for faster analyst review, trend exploration, and downstream reporting.

Baseball Statcast Pipeline

Python, Flask REST API, SQL

- Engineered a player-performance analytics pipeline for xwOBA, Stuff+, and sprint speed metrics with reusable REST API endpoints.

Portfolio Analytics Platform

Monte Carlo, VaR/CVaR, Flask

- Developed a risk analytics app that simulates portfolio outcomes, estimates VaR/CVaR, and exposes results through Flask.

Baltimore Crime Analytics

SQL, GeoPandas, Tableau

- Analyzed neighborhood-level crime patterns and built geospatial dashboards for trend comparison and public-safety exploration.

Publications

Leveraging Fog Computing, Blockchain, and IoT for Enhanced Healthcare Monitoring, IJRASET, 2024. DOI: 10.22214/ijraset.2024.58671

Enhanced Anomaly Detection Framework for 6G Software-Defined Networks, IJISRT, 2024